Organic Cume, September 24, 2021

1. Draw the lowest energy conformation for *cis*-1,4-dimethylcyclohexane.

2. Why is propylcyclopropane (C_6H_{12}) higher in potential energy compared to cyclohexane (C_6H_{12})?

3. Draw two C_5H_{10} constitutional isomers.

4. Draw two C_5H_{10} stereoisomers.

5. What is the ee of a mixture of enantiomers that having an er = 80:20?

6. Hydrolysis (ethanol/H_2O) of (*R*)-2-bromoooctane gives (*S*)-2-octanol and (*R*)-2-octanol in 18% and 17%, respectively. Explain.

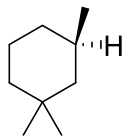
6. Give the mechanism for the hydrolysis reaction of 2-bromo-3-methylbutane to give 2-methyl-2-butanol.

8. For many S_N2 reactions, aprotic solvents are preferred over protic solvents. Contrast the reaction of methylbromide reacting with methoxide in both solvent systems using reaction coordinate diagrams.

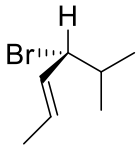
9. Structurally, what makes *p*-toluenesulfonate a good leaving group for S_N2 reactions?

10. Give the mechanism and products for the reaction of 2-butanol with thionyl chloride and pyridine.

1.



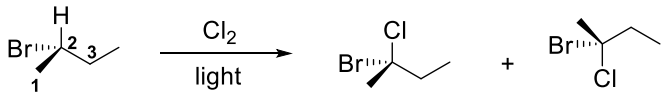
A



B

(*R*)-1,1,3-trimethylcyclohexane (*R*,*E*)-4-bromo-5-methylhex-2-ene

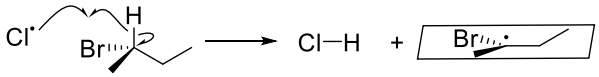
2.



enantiomers formed in equal amounts

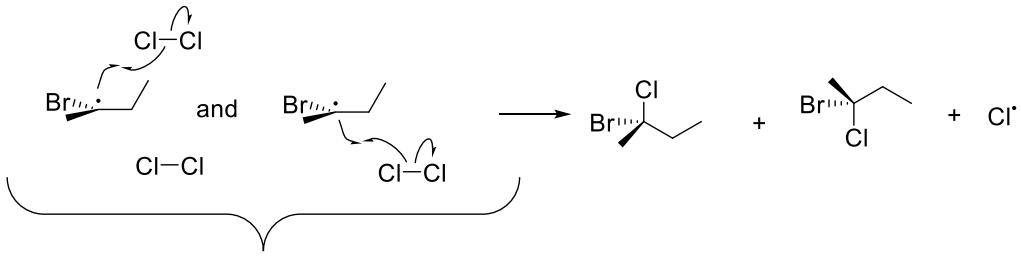
Propagation steps:

Step 1



plane of symmetry

Step 2



transition states from this step are enantiomeric so have identical activation energies, thus giving equal amounts of products

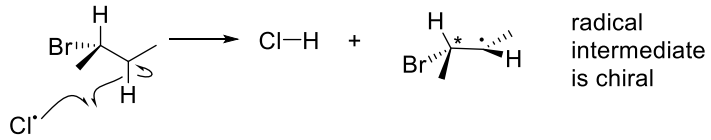
[

okay to use a radical intermediate that is slightly pyramidal and adjust argument appropriately]

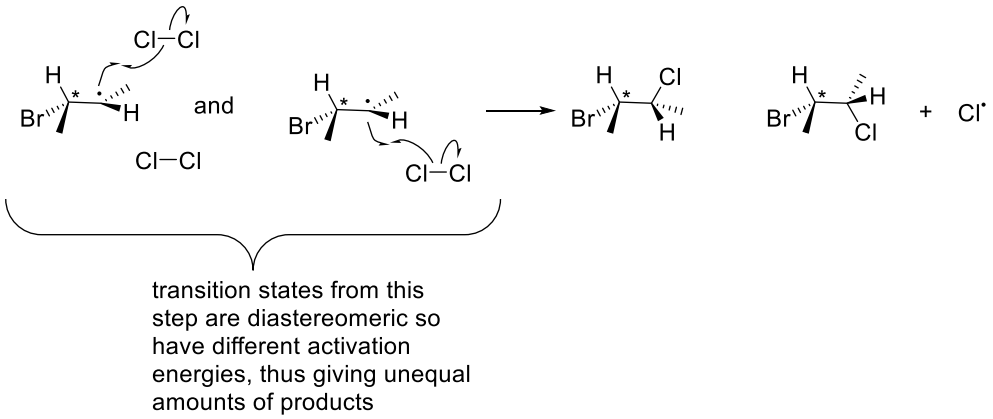
3.

Propagation steps:

Step 1



Step 2



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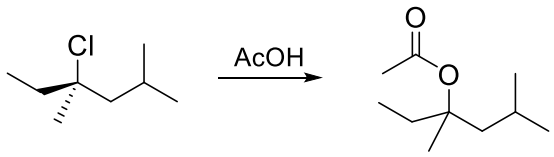

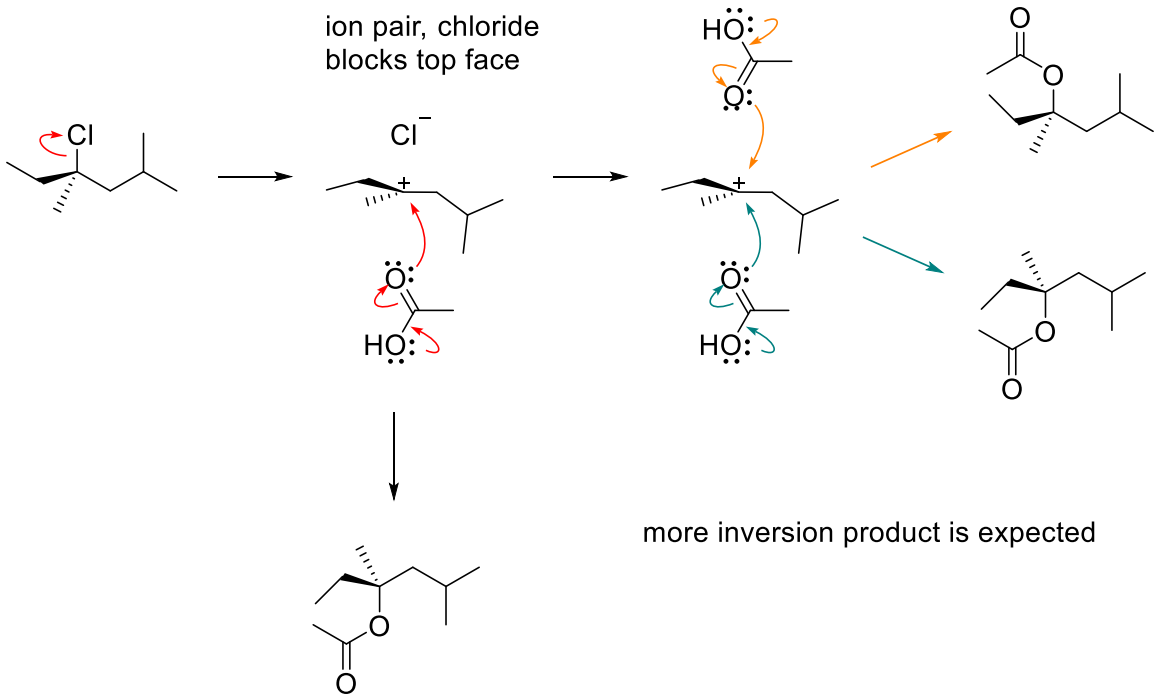

more inversion product is expected